NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

NTU Academy for Professional and Continuing Education

How LLMs Work

Looking into the inner workings of LLMs

Topic 4

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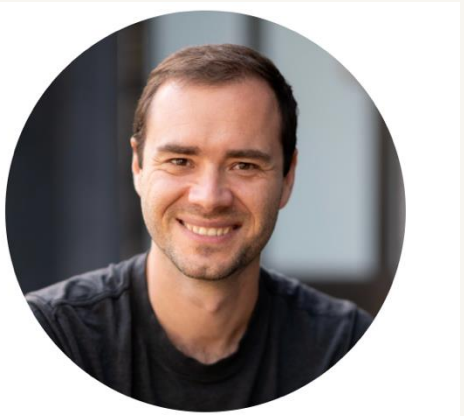
Karpathy's micro-gpt



Andrej Karpathy

https://github.com/karpathy

microgpt

The most atomic way to train and inference a GPT in pure, dependency-free Python. This file is the complete algorithm. Everything else is just efficiency.

```
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The most atomic way to train and inference a GPT in pure, dependency-free Python.
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@karpathy
"""

import os       # os.path.exists
import math     # math.log, math.exp
import random   # random.seed, random.choices, random.gauss, random.shuffle
random.seed(42) # Let there be order among chaos

# Let there be an input dataset `docs`: list[str] of documents (e.g. a dataset of names)
if not os.path.exists('input.txt'):
    import urllib.request
    names_url = 'https://raw.githubusercontent.com/karpathy/makemore/refs/heads/master/names.txt'
    urllib.request.urlretrieve(names_url, 'input.txt')
docs = [l.strip() for l in open('input.txt').read().strip().split('\n') if l.strip()] # list[str] of documents
random.shuffle(docs)
print(f"num docs: {len(docs)}")

# Let there be a Tokenizer to translate strings to discrete symbols and back
uchars = sorted(set(''.join(docs))) # unique characters in the dataset become token ids 0..n-1
BOS = len(uchars) # token id for the special Beginning of Sequence (BOS) token
vocab_size = len(uchars) + 1 # total number of unique tokens, +1 is for BOS
print(f"vocab size: {vocab_size}")

# Let there be an Autograd to apply the chain rule recursively across a computation graph
class Value:
    """Stores a single scalar value and its gradient, as a node in a computation graph."""

    def __init__(self, data, children=(), local_grads=()):
        self.data = data                # scalar value of this node calculated during forward pass
        self.grad = 0                   # derivative of the loss w.r.t. this node, calculated in backward pass
        self._children = children       # children of this node in the computation graph
        self._local_grads = local_grads # local derivative of this node w.r.t. its children

    def __add__(self, other):
        other = other if isinstance(other, Value) else Value(other)
        return Value(self.data + other.data, (self, other), (1, 1))

    def __mul__(self, other):
        other = other if isinstance(other, Value) else Value(other)
        return Value(self.data * other.data, (self, other), (other.data, self.data))

    def __pow__(self, other): return Value(self.data**other, (self,), (other * self.data**(other-1),))
    def log(self): return Value(math.log(self.data), (self,), (1/self.data,))
    def exp(self): return Value(math.exp(self.data), (self,), (math.exp(self.data),))
    def relu(self): return Value(max(0, self.data), (self,), (float(self.data > 0),))
    def __neg__(self): return self * -1
    def __radd__(self, other): return self + other
    def __sub__(self, other): return self + (-other)
    def __rsub__(self, other): return other + (-self)
    def __rmul__(self, other): return self * other
    def __truediv__(self, other): return self * other**-1
    def __rtruediv__(self, other): return other * self**-1

    def backward(self):
        topo = []
        visited = set()
        def build_topo(v):
            if v not in visited:
                visited.add(v)
                for child in v._children:
                    build_topo(child)
                topo.append(v)
        build_topo(self)
        self.grad = 1
        for v in reversed(topo):
            for child, local_grad in zip(v._children, v._local_grads):
                child.grad += local_grad * v.grad

# Initialize the parameters, to store the knowledge of the model.
n_embd = 16     # embedding dimension
n_head = 4      # number of attention heads
n_layer = 1     # number of layers
block_size = 8  # maximum sequence length
head_dim = n_embd // n_head # dimension of each head
matrix = lambda nout, nin, std=0.02: [[Value(random.gauss(0, std)) for _ in range(nin)] for _ in range(nout)]
state_dict = {'wte': matrix(vocab_size, n_embd), 'wpe': matrix(block_size, n_embd), 'lm_head': matrix(vocab_size, n_embd)}
for i in range(n_layer):
    state_dict[f'layer{i}.attn_wq'] = matrix(n_embd, n_embd)
    state_dict[f'layer{i}.attn_wk'] = matrix(n_embd, n_embd)
    state_dict[f'layer{i}.attn_wv'] = matrix(n_embd, n_embd)
    state_dict[f'layer{i}.attn_wo'] = matrix(n_embd, n_embd, std=0)
    state_dict[f'layer{i}.mlp_fc1'] = matrix(4 * n_embd, n_embd)
    state_dict[f'layer{i}.mlp_fc2'] = matrix(n_embd, 4 * n_embd, std=0)
params = [p for mat in state_dict.values() for row in mat for p in row] # flatten params into a single list[Value]
print(f"num params: {len(params)}")

# Define the model architecture: a stateless function mapping token sequence and parameters to logits over what comes next.
# Follow GPT-2, blessed among the GPTs, with minor differences: layernorm -> rmsnorm, no biases, GeLU -> ReLU^2
def linear(x, w):
    return [sum(wi * xi for wi, xi in zip(wo, x)) for wo in w]

def softmax(logits):
    max_val = max(val.data for val in logits)
    exps = [(val - max_val).exp() for val in logits]
    total = sum(exps)
    return [e / total for e in exps]

def rmsnorm(x):
    ms = sum(xi * xi for xi in x) / len(x)
    scale = (ms + 1e-5) ** -0.5
    return [xi * scale for xi in x]

def gpt(token_id, pos_id, keys, values):
    tok_emb = state_dict['wte'][token_id] # token embedding
    pos_emb = state_dict['wpe'][pos_id] # position embedding
    x = [t + p for t, p in zip(tok_emb, pos_emb)] # joint token and position embedding
    x = rmsnorm(x)

    for li in range(n_layer):
        # 1) Multi-head attention block
        x_residual = x
        x = rmsnorm(x)
        q = linear(x, state_dict[f'layer{li}.attn_wq'])
        k = linear(x, state_dict[f'layer{li}.attn_wk'])
        v = linear(x, state_dict[f'layer{li}.attn_wv'])
        keys[li].append(k)
        values[li].append(v)
        x_attn = []
        for h in range(n_head):
            hs = h * head_dim
            q_h = q[hs:hs+head_dim]
            k_h = [ki[hs:hs+head_dim] for ki in keys[li]]
            v_h = [vi[hs:hs+head_dim] for vi in values[li]]
            attn_logits = [sum(q_h[j] * k_h[t][j] for j in range(head_dim)) / head_dim**0.5 for t in range(len(k_h))]
            attn_weights = softmax(attn_logits)
            head_out = [sum(attn_weights[t] * v_h[t][j] for t in range(len(v_h))) for j in range(head_dim)]
            x_attn.extend(head_out)
        x = linear(x_attn, state_dict[f'layer{li}.attn_wo'])
        x = [a + b for a, b in zip(x, x_residual)]
        # 2) MLP block
        x_residual = x
        x = rmsnorm(x)
        x = linear(x, state_dict[f'layer{li}.mlp_fc1'])
        x = [xi.relu() ** 2 for xi in x]
        x = linear(x, state_dict[f'layer{li}.mlp_fc2'])
        x = [a + b for a, b in zip(x, x_residual)]

    logits = linear(x, state_dict['lm_head'])
    return logits

# Let there be Adam, the blessed optimizer and its buffers
learning_rate, beta1, beta2, eps_adam = 1e-2, 0.9, 0.95, 1e-8
m = [0.0] * len(params) # first moment buffer
v = [0.0] * len(params) # second moment buffer

# Repeat in sequence
num_steps = 500 # number of training steps
for step in range(num_steps):

    # Take single document, tokenize it, surround it with BOS special token (token id 0) on both sides
    doc = docs[step % len(docs)]
    tokens = [BOS] + [uchars.index(ch) for ch in doc] + [BOS]
    n = min(block_size, len(tokens) - 1)

    # Forward the token sequence through the model, building up the computation graph all the way to the loss.
    keys, values = [[] for _ in range(n_layer)], [[] for _ in range(n_layer)]
    losses = []
    for pos_id in range(n):
        token_id, target_id = tokens[pos_id], tokens[pos_id + 1]
        logits = gpt(token_id, pos_id, keys, values)
        probs = softmax(logits)
        loss_t = -probs[target_id].log()
        losses.append(loss_t)
    loss = (1 / n) * sum(losses) # final average loss over the document sequence. May yours be low.

    # Backward the loss, calculating the gradients with respect to all model parameters.
    loss.backward()

    # Adam optimizer update: update the model parameters based on the corresponding gradients.
    lr_t = learning_rate * 0.5 * (1 + math.cos(math.pi * step / num_steps)) # cosine learning rate decay
    for i, p in enumerate(params):
        m[i] = beta1 * m[i] + (1 - beta1) * p.grad
        v[i] = beta2 * v[i] + (1 - beta2) * p.grad ** 2
        m_hat = m[i] / (1 - beta1 ** (step + 1))
        v_hat = v[i] / (1 - beta2 ** (step + 1))
        p.data -= lr_t * m_hat / (v_hat ** 0.5 + eps_adam)
        p.grad = 0

    print(f"step {step+1:4d} / {num_steps:4d} | loss {loss.data:.4f}")

# Inference: may the model babble back to us
temperature = 0.5 # in (0, 1], control the "creativity" of generated text, low to high
print("\n--- inference ---")
for sample_idx in range(20):
    keys, values = [[] for _ in range(n_layer)], [[] for _ in range(n_layer)]
    token_id = BOS
    sample = []
    for pos_id in range(block_size):
        logits = gpt(token_id, pos_id, keys, values)
        probs = softmax([l / temperature for l in logits])
        token_id = random.choices(range(vocab_size), weights=[p.data for p in probs])[0]
        if token_id == BOS:
            break
        sample.append(uchars[token_id])
    print(f"sample {sample_idx+1:2d}: {''.join(sample)}")
```

@karpathy

Karpathy's micro-gpt

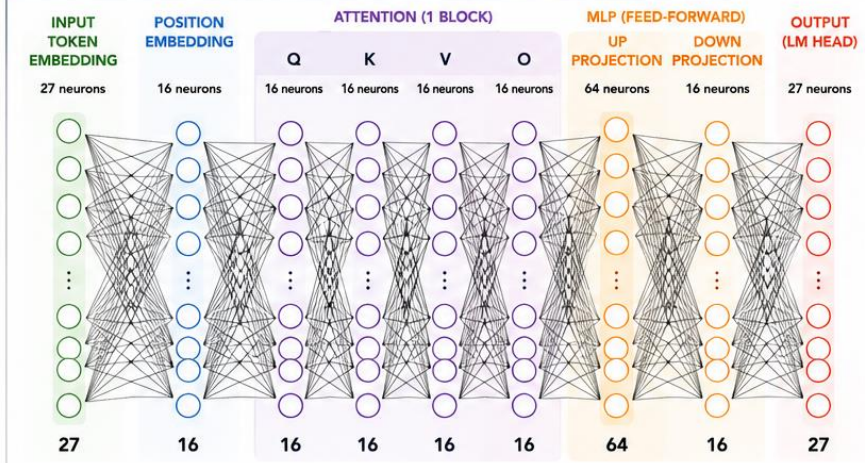
4,192 PARAMETERS = 4,192 LINES = 4,192 DIALS TO TURN

A neural network is a drawing of circles connected by lines. The circles are neurons. The lines between them are weights — each line carries one number that says "how strongly does the signal on this line flow into the next neuron." The model is those lines. Each line is a dial — one number we can adjust.

Training is the process of turning every one of those 4,192 dials, one tiny step at a time, until the lines collectively carry the signal in a way that produces useful predictions.

So "4,192 parameters" = "4,192 lines in the diagram" = "4,192 dials to turn" = "4,192 points of adjustment to reduce loss." Four ways of saying the same thing.

MICROGPT AS A DRAWING OF NEURONS AND WEIGHTS



Neuron (a unit that holds a value)

Weight (one number on a connection)

Every line has one number (a weight). That number is a dial we can turn.

WHY THE COUNT IS EXACTLY 4,192

1	27 → 16 at the input (token embedding)	27 × 16 = 432
2	16 → 16 for position embedding	16 × 16 = 256
3	16 → 16 four times in attention (Q, K, V, O)	4 × 256 = 1,024
4	16 → 64 in the MLP up-projection	16 × 64 = 1,024
5	64 → 16 in the MLP down-projection	64 × 16 = 1,024
6	16 → 27 at the output (lm_head)	16 × 27 = 432
	TOTAL	**4,192 LINES**

THE BIG PICTURE

The whole machine is just lines connecting neurons.
Training is the act of finding the right setting for every line.

When you hear "GPT-4 has hundreds of billions of parameters," it means the same thing at a vastly bigger scale: hundreds of billions of lines drawn between neurons, each one a dial the optimiser turned during training.

Karpathy's micro-gpt

```
docs = [...]                        # load names
uchars = sorted(set(''.join(docs)))    # build vocabulary

state_dict = {'wte': ..., 'wpe': ...}  # initialise a big dict of matrices of Values

for step in range(num_steps):         # 1000 iterations
   # forward pass through gpt(), compute loss, loss.backward(), update params

for sample_idx in range(20):          # generate 20 samples
   # call gpt() repeatedly, sample a token, append to output
```

Resources

How LLM Work - Visual Explanation	https://www.youtube.com/watch?v=LPZh9BOjkQs
Interactive App – Playing with micro-gpt	https://microgpt-leaders.vercel.app/
Deep Dive into LLMs like ChatGPT - Karpathy	https://www.youtube.com/watch?v=7xTGNNLPyMI
Demo Code Repo 1	https://github.com/SwarupSG/micro-gpt-demo-code
Demo Code Repo 2	https://github.com/SwarupSG/how-llm-work-microgpt

Version Management

Month/Year	Version	Changes
May 2026	0.0	New Module Design